Mishkat Noor

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Linkedin

My work is driven by the synergy between computational science and biology, with a specific focus on developing pragmatic solutions for the medical field. I am particularly interested in advancing the scope of neuro-bioinformatics by leveraging artificial intelligence and machine learning to interpret complex biological data. Furthermore, I am committed to refining my scientific communication through rigorous research and peer-reviewed publications, ensuring that my approach to bioinformatics remains both creative and methodologically sound.

EDUCATION

- **National University of Science and Technology (NUST)**
  BS Bioinformatics (**2023-27)**

ACADEMIC COURSEWORK

- Molecular Biology
- Evolutionry Biology
- Genetics
- Bioinformatics
- NGS
- Discrete Mathematics
- Linear Algebra
- Calculus I and II
- Data Structures and Algo.
- Object Oriented Programming
- AI and ML
- Database Management Sys
- Digital Logic Design
- Computer Organization and Assembly Languages
- Technical Writing and Expository Writing
- Professional Ethics and Management
- Software Engineering

PROJECTS

1. **End-to-End Long-Read Variant Calling & Benchmarking Pipeline on HPC (HG002)**

Focus Area: Bioinformatics / NGS / High-Performance Computing
Skills: Nextflow workflow development · SLURM job scheduling · Singularity containerization · Long-read alignment (Minimap2) · BAM processing (Samtools) · Deep learning variant calling (Clair3, DeepVariant) · GRCh38 reference handling · .fai indexing · GIAB truth set benchmarking · hap.py performance evaluation · Precision/Recall/F1 analysis · Reproducible research pipeline design · Linux & HPC environment management.

- Designed and implemented a production-ready, containerized whole-genome variant calling pipeline for PacBio HiFi HG002 data on an HPC cluster using SLURM, Nextflow, and Singularity. The workflow included long-read alignment to GRCh38, BAM sorting and indexing, parallel variant calling using Clair3 and DeepVariant, and benchmarking against the GIAB v4.2.1 truth set using hap.py. The project demonstrated applied competency in scalable workflow orchestration, reference genome management, deep learning-based variant detection, and quantitative performance evaluation (Precision, Recall, F1-Score) within a reproducible bioinformatics framework.

2. **Replication of Vertebrate Genome Project (VGP) Assembly Pipeline**

Focus Area: Bioinformatics / Genome Assembly / NGS Workflow Development
Skills: De novo genome assembly · Long-read data processing · Assembly polishing · Quality assessment · Structural evaluation · HiFi & ONT reads handling · Contig/scaffold generation · BUSCO and QUAST evaluation · Galaxy workflows · Reproducible analytic pipelines · GitHub documentation · HPC usage

- The project involved replicating the Vertebrate Genome Project (VGP) assembly pipeline using publicly available sequencing data and the Galaxy platform training materials. Key tasks included long-read assembly generation, iterative polishing, scaffolding, assembly validation, and quality assessment using tools such as Minimap2, Flye, Racon, Medaka, and BUSCO. The workflow produced a high-quality draft assembly for a model organism, with comprehensive metrics generated via QUAST and BUSCO to evaluate completeness and contiguity. The implementation emphasized modular workflow design, rigorous quality control, and documentation to enable reproducible large-genome assembly in a research environment.

3. **Curation and Standardization of Clinically Significant Genetic Variants into GRCh38 VCF and further annotating it**

Focus Area: Bioinformatics/NGS.
Skills: Variant interpretation and curation. GRCh38 genomic coordinate analysis . VCF (Variant Call Format) generation and formatting . HGVS nomenclature interpretation. UCSC Genome Browser usage . structural and SNV variant classification . Genomic data standardization.

- The project involved extracting genomic positions, interpreting HGVS nomenclature, and generating a standards-compliant VCF (v4.2) file representing multiple variant types for downstream analysis. The VCF was

further annotated using bamtools with ClinVar, demonstrating applied competency in variant interpretation, genomic coordinates, and bioinformatics standards.

4. Finding Similarities And Evolutionary Relationships Through Protein Sequences

Focus Area: Bioinformatics / Molecular Evolution, Third Semester.
Skills: Molecular Phylogeny · Molecular Evolution · Jalview · MEGA11 · iTol.

- Analyzed protein sequence similarities and evolutionary relationships across plants, fungi, and animals using FASTA and Jalview for phylogenetic visualization.
- Gained hands-on experience in sequence alignment, phylogenetic interpretation, and comparative genomics.

5. Neuro AI – Predictive Stress-Based Disease Modeling

Focus Area: Neuro-Bioinformatics, Mental Health AI
Skills: Python for ML · Artificial Neural Networks · Streamlit · UI · Research Skills

- Developed Neuro AI, a clinical research backed predictive tool integrating neurobiology, psychology, and machine learning to assess susceptibility to stress induced chronic diseases.

6. Computational Epidemiology for Pandemic Forecasting

Focus Area: Bioinformatics, AI in Public Health, Third Semester.
Skills: SEIR · Panedemic Epidimology

- Studied the role of computational epidemiology in pandemic forecasting and management using existing COVID-19 datasets.
- Explored how bioinformatics tools (Nextstrain, GISAID) and AI models enable mutation tracking, phylogenetic mapping, and outbreak prediction.

7. Molecular Modeling of Axonal Lipid Membrane using CHARMM-GUI

Focus Area: Computational Cell Biology / Structural Bioinformatics, Second Semester.
Skills: Research Skills · Research Writing · Molecular Modeling · Biochemistry · Bioinformatics · CHARMM

- Simulated the lipid membrane composition of a neuronal axon using CHARMM-GUI, constructing a biologically relevant bilayer to study membrane structure and dynamics.
- Customized lipid types, built the model, and prepared the system for molecular dynamics simulations.

8. Cambridge Analytica Scandal – Ethics in Data Privacy

Focus Area: Ethics for Bioinformatics.
Skills: Professional Ethics · Data Privacy Awareness · Presentation Skills · Public Speaking · Public Strategy

- Examined the Cambridge Analytica scandal as a case study in data ethics, focusing on privacy breaches, user consent violations, and political manipulation.
- Analyzed ethical implications of data mining, corporate responsibility, and the need for stronger global data ethics frameworks.

9. Social Media Strategy for Verstees – A Practical Brand Development

Focus Area: Management Sciences.
Skills: Social Media Marketing, Optimization and Strategy · Project Management · Corporate Communications.

- Developed a social media strategy for Verstees, a fashion/lifestyle brand, applying management principles to branding, digital engagement, and audience growth.
- Designed content calendars, competitor analysis, influencer strategies, and KPIs to translate theory into practical marketing outcomes.

10. Virtual Doctor App

Focus Area: Computer Programming and fundamentals, Second Semester.
Skills: C (Programming Language) · Problem Solving · Teamwork

- Built an individual Virtual Doctor App in C as part of an end-semester project to distinguish between dengue and malaria based on patient symptoms.
- Strengthened core programming skills and gained confidence in developing medical diagnostic tools.

CERTIFICATIONS & TRAININGS

- AWS Cloud Practitioner Training
- Git Ops
- Lapel Pin for Best Club Office Bearers of the year (2024-25)

LEADERSHIP/EXTRACURRICULAR

- Deputy Director Social Media Management (SMM)
- NUST Social Values Society (NSVS) (2025-) Press Secretary
- Nust Quiz Club (NQC), (2024-25)
- School Orientation Leader (SOL), Orientation of NUST (2024).
- Director Publications, Nust Quiz Club(NQC) (2023-24).